Domain Name System (DNS) Architecture

Coverage: Hierarchical Distributed Database, Root/TLD/Authoritative Servers, Iterative vs Recursive Queries, Resource Records (RR)

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1. DNS Hierarchy & Resolution

- Distributed Database translating human hostnames to IP addresses over UDP port 53.
- Hierarchy: Root Servers -> TLD Servers (.com, .in) -> Authoritative Servers (vjcet.ac.in) -> Local Resolvers.
- Iterative: Contacted server returns referral. Recursive: Contacted server resolves full IP on client behalf.

2. Resource Records (RR)

- Type A (IPv4), Type AAAA (IPv6), Type CNAME (Alias), Type NS (Name server), Type MX (Mail server), Type PTR (Reverse lookup).

■ KTU University High-Yield Exam Questions

- Q: Draw and explain the step-by-step resolution process for a domain using Iterative vs Recursive queries. (8 Marks)